

Subject

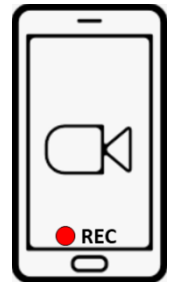
Speech and Natural Language Processing

Professor

Isaac González



Aula Virtual U/E



Participar



Session 01

Unit I

Natural Language Processing

Topic 1

Introduction to NLP

Index

- The subject
- Me
- You
- Debate
- Philosophy of the classes
- Goal

Natural Language Processing

Wikipedia:

“Natural language processing (NLP) is a subfield of computer science and especially artificial intelligence. It is primarily concerned with providing computers with the ability to **process data encoded in natural language** and is thus closely related to information retrieval, knowledge representation and computational linguistics, a subfield of linguistics. Typically data is collected in **text corpora, using either rule-based, statistical or neural-based approaches in machine learning and deep learning**. Major tasks in natural language processing are speech recognition, text classification, natural-language understanding, and natural-language generation.”

Natural Language Processing

My vision:

“Natural Language processing is the computer’s use of the human language (natural language) in any sense.”

Course sectioning:

1. NLP the old way: rule-based, understand the problems in managing natural language with computers
2. Deep learning for NLP: How neural networks changed NLP
3. Transformers and LLMs
4. Speech processing

Me, my story

Isaac González

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My story:

- Computer Science Engineer
- NLP Researcher
- MBA
- NLP Spin-off Founder and CEO
- PhD Student in Blockchain and DeFi

Me, my story

> Software is eating the world.

Marc Andreessen (A16Z VC) at 2011

<https://a16z.com/why-software-is-eating-the-world/>

> Why Bitcoin matters

Marc Andreessen (A16Z VC) at 2014

<https://a16z.com/why-bitcoin-matters/>

> Why AI will save the world.

Marc Andreessen (A16Z VC) at 2023

<https://a16z.com/ai-will-save-the-world/>

You, your story, and your future

We are going to spend some time together. The most effective way to learn is when I know you.

- Name
- Where did you come from?
- Why do you choose this career
- Your hobbies, in what activities apart from the university you spend your time?
- What is the most exciting thing in the world for you
- What do you want to do after graduation? (Work, I know, but give me details of your desires)
- How do you imagine yourself in 15 years? What would be your ideal life?

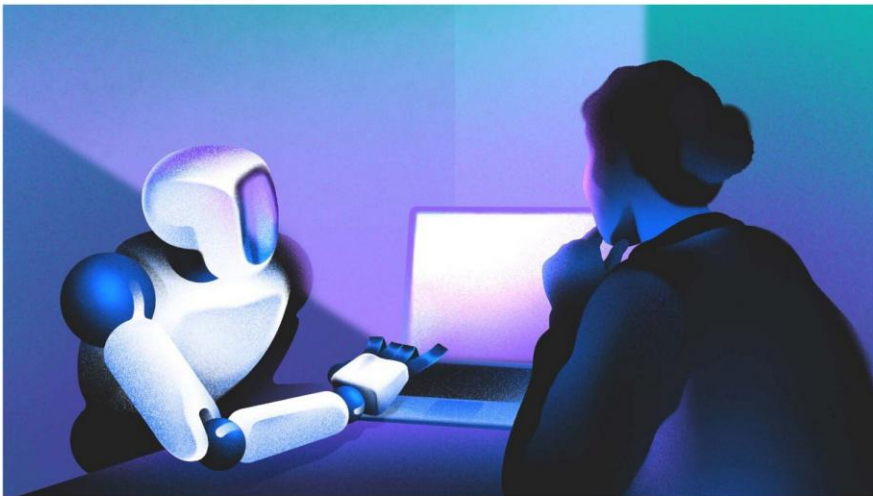
Debate: The junior dilemma in AI era

- Tech jobs are shrinking because of AI
- Juniors are not hired because AI can do simple to medium tasks
- Juniors do not expend the time to become seniors
- What are you going to do?

ARTIFICIAL INTELLIGENCE • WORKPLACE

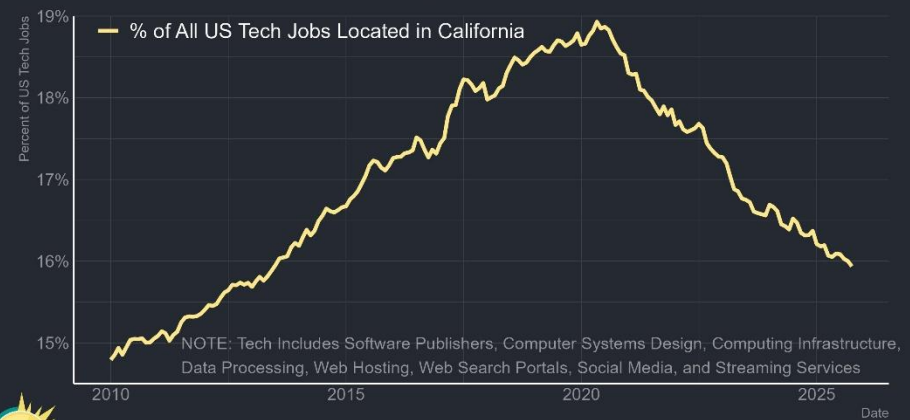
The AI Vs. Junior Talent Dilemma

Itay Sagie August 29, 2025



Tech Jobs are Leaving California

California's Share of Total Tech-Sector Employment Has Fallen Significantly Since 2020



Graph created by @JosephPolitano using BLS data

Course philosophy

- NLP is a huge topic! It is changing a lot!
- The course tends to give a global overview of concepts, mixing programming algorithms, using libraries, and using APIs.
- The idea is to give you the tools to understand the topic in a broad sense
- You should be very proactive and lean into experimentation and own research
- Your portfolio speaks about you; take care of what it says.

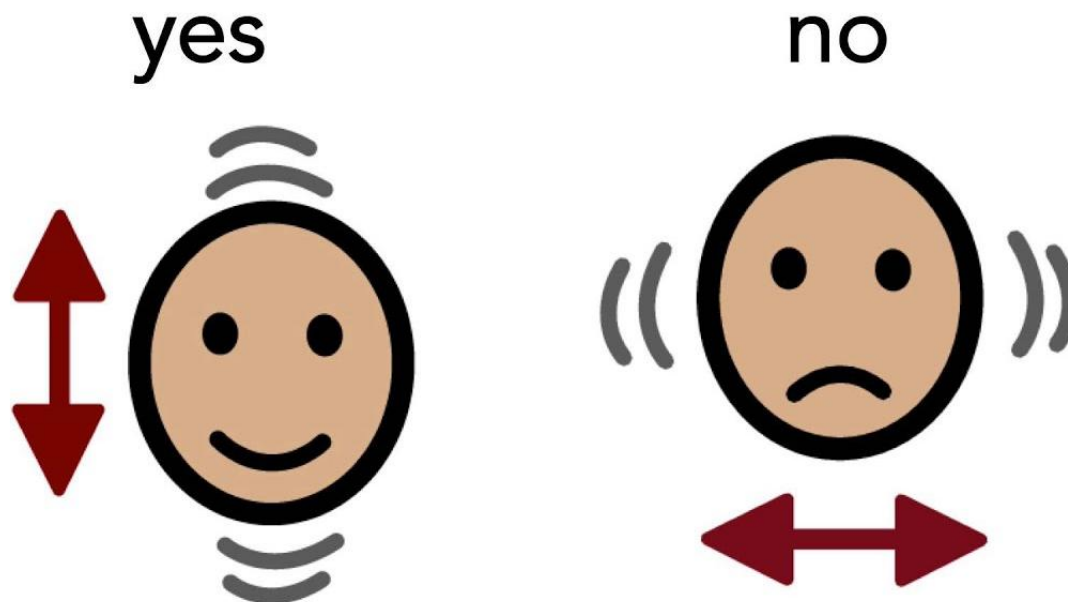
Course philosophy

Questions? Yes, ask questions!

- It is crucial you gain an active role in the relation with questions, interests, problems, or exciting things you discover.

Participation

- Since this subject is in English, it is a complex matter. Please let me know when you get it, or if you don't.
- If you are shy to say, you can move your head like that.



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Honor Code

- No cheating or plagiarism is allowed.
- **LLMs are not forbidden by default, but you should understand what you stand for.**
- Do not fool me, it is not good for us, especially for your honor.

Next week, we will see the
Teaching-Learning agreement document

Y

Software in the era of AI

Andrej Karpathy

Founding Member, OpenAI

Digital Portfolio = GitHub repository

Git, github, and VSCode recap

- Git: version control system
 - Install git <https://git-scm.com/book/en/v2/Getting-Started-Installing-Git>
 - Book <https://git-scm.com/book/en/v2>
- Github: git server + Project management (and more!)
 - Getting started
 - Create an account on github.com
- VSCode + git + github
 - VSCode is the go-to IDE to interact with git and GitHub
 - Link VSCode and Your GitHub account: Staging and committing
 - Pushing and pulling
- Markdown
 - [Github Markdown](#)

Python and Notebooks

Let's check what you know!

1. Create in github your portfolio (public or private shared with me: "isaacgonzalez"). Name: "NLP_portfolio_surname_name".
2. Clone the repository in your local environment
3. Install the extensions of VSCode to execute Python Notebooks
4. Download from the Campus Virtual the file "S01_exercises.ipynb"
5. Do the exercises **WITHOUT LLMs help**, only you and the manual.
6. Check with the teacher any doubt of the process or the exercises.

Exercises after class

- Finish S01_exercises.ipynb
- Do **AE05 Forum**: AI and language in commercial webs and apps